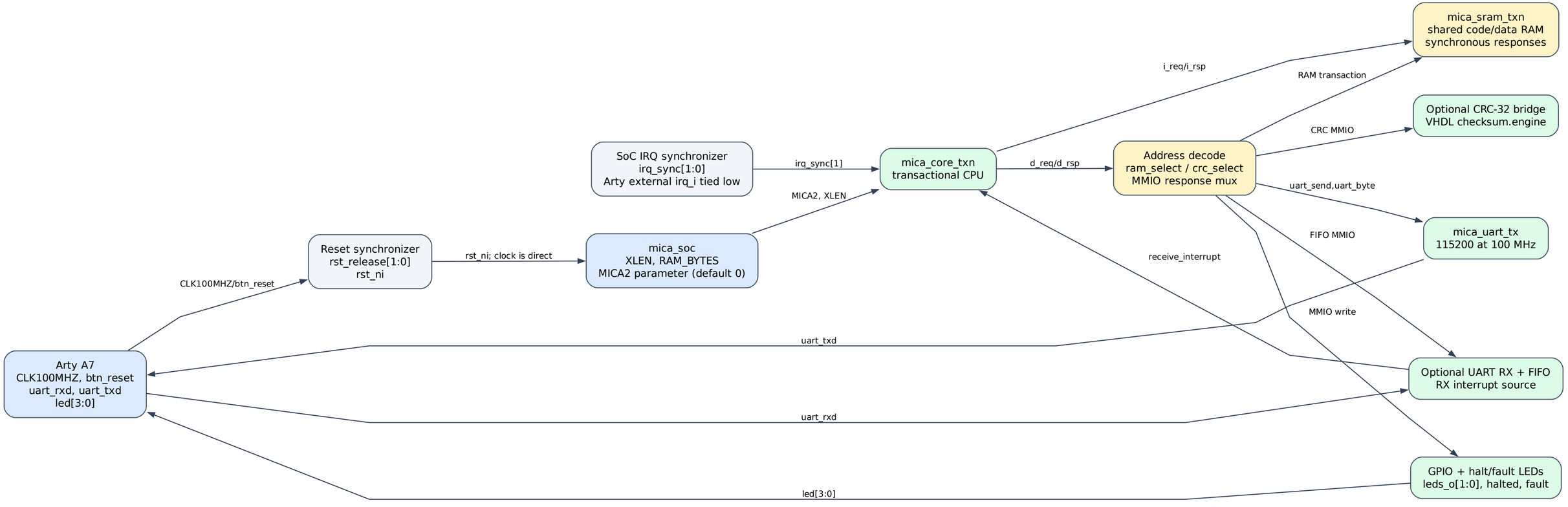
MICA 2.0 | SoC wiring and board-level boundary
mica_sram_txn
shared code/data RAM
synchronous responses
Optional CRC-32 bridge
VHDL checksum.engine
mica_uart_tx
115200 at 100 MHz
Optional UART RX + FIFO
RX interrupt source
GPIO + halt/fault LEDs
leds_o[1:0], halted, fault
SoC IRQ synchronizer
irq_sync[1:0]
Arty external irq_i tied low
mica_core_txn
transactional CPU
Address decode
ram_select / crc_select
MMIO response mux
Reset synchronizer
rst_release[1:0]
rst_ni
mica_soc
XLEN, RAM_BYTES
MICA2 parameter (default 0)
Arty A7
CLK100MHZ, btn_reset
uart_rxd, uart_txd
led[3:0]
irq_sync[1]
d_req/d_rsp
i_req/i_rsp
RAM transaction
CRC MMIO
uart_send,uart_byte
FIFO MMIO
receive_interrupt
MMIO write
MICA2, XLEN
rst_ni; clock is direct
CLK100MHZ/btn_reset
uart_txd
uart_rxd
led[3:0]